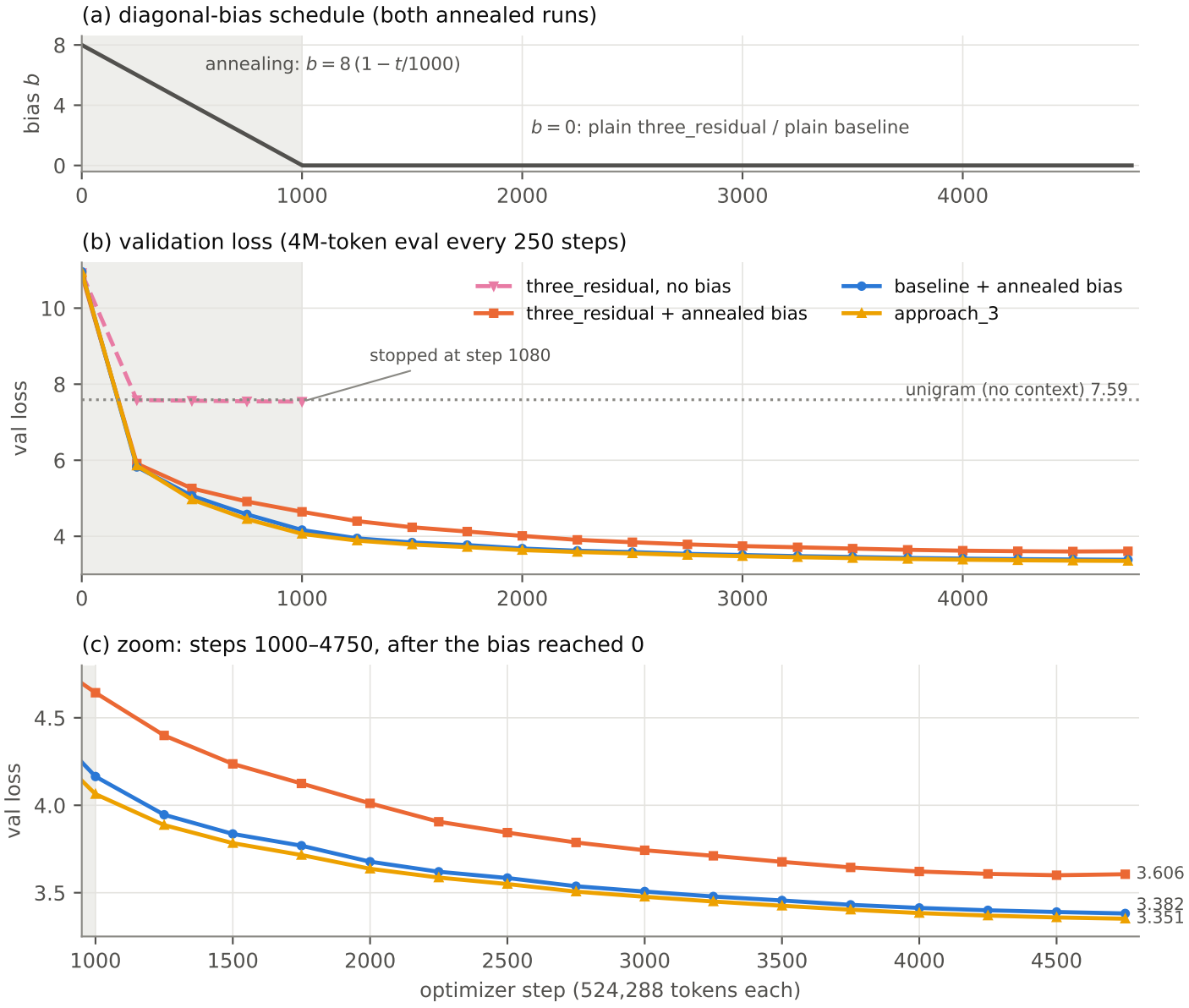
(a) diagonal-bias schedule (both annealed runs)
bias b
annealing: b = 8 (1 − t/1000)
b = 0: plain three_residual / plain baseline
(b) validation loss (4M-token eval every 250 steps)
val loss
three_residual, no bias
three_residual + annealed bias
baseline + annealed bias
approach_3
stopped at step 1080
unigram (no context) 7.59
(c) zoom: steps 1000–4750, after the bias reached 0
val loss
3.606
3.382
3.351
optimizer step (524,288 tokens each)